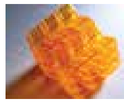
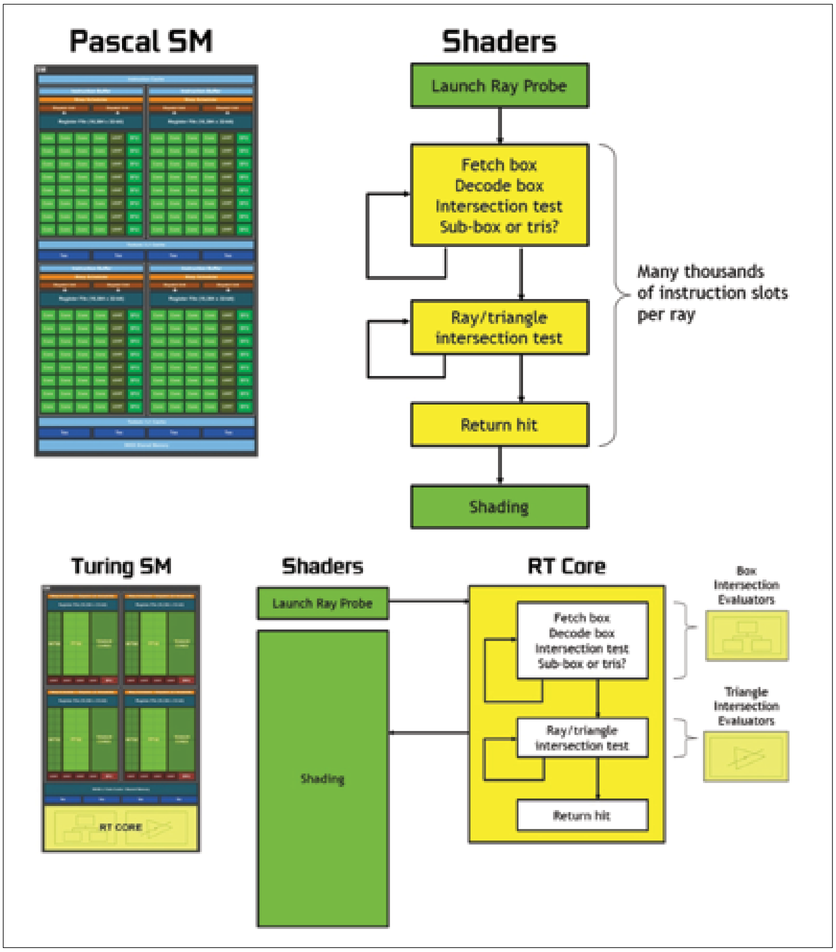
"Falling stars" challenge
Faceplanting while surrounded by luxury items is the latest fad for rich kids on social networks. **https://dgit.in/nov18-18-3**



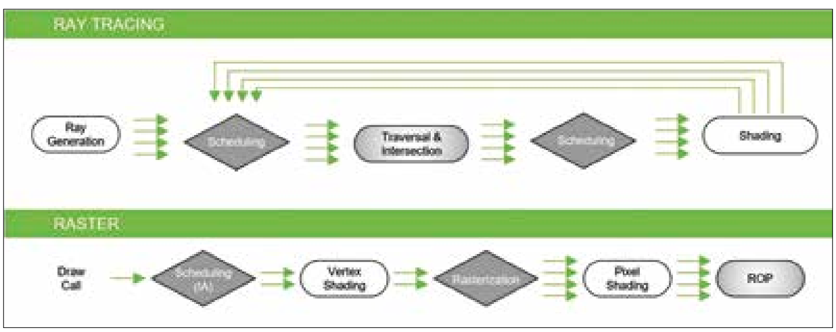
Origami 3D printing
Researchers have merged the ancient art of origami with 3D printing to build complex structures easily. **https://dgit.in/nov18-18-2**



Comparison of Pascal and Turing shaders

Think of it as a trade-off. While tracing each and every ray of light per pixel can obviously result in greater realism, there is the rule of diminishing returns at play. Firstly, we are looking at a 2-dimensional display and there are inherent limitations of the display panel as well. So if the display panel cannot reproduce the outrageous levels of realism that can be achieved, why bother then? Might as well scale it down a notch till the viewer cannot perceive a loss in fidelity.

NVIDIA's Turing GPUs accelerate ray tracing by using a combination of several techniques. Some of which will be familiar, while some will be new. Not all of these techniques come into play during the rendering stage. Regardless, here's everything that Turing makes use of to achieve real-time ray tracing.



Hybrid rendering pipeline

- Reflections and Refractions
- Shadows and Ambient Occlusion
- Global Illumination
- Instant and off-line lightmap baking
- Beauty shots and high-quality previews
- Primary rays for foveated VR rendering
- Occlusion Culling
- Physics, Collision Detection, Particle simulations
- Audio simulation
- AI visibility queries
- In-engine Path Tracing

The RT cores accelerate Bounding Volume Hierarchy (BVH) traversal and ray/triangle intersection testing. These two processes need to be performed in an iterative fashion as the BVH traversal would need thousands of intersection testing to finally calculate the colour of the pixels. Since RT cores are specialised to take on this load, this gives the shader cores in the Streaming Multiprocessor (a sub-section of the GPU) room to focus on other aspects of the scene.

Without delving too much into the numbers, NVIDIA's whitepaper states that Pascal GPU can handle 1.1 Giga Rays/Sec while Turing can do more than 10 Giga Rays/Sec. This is because of how specialised the RT cores are and due to the hybrid rendering pipeline used by the NVIDIA Turing architecture.

In order to attain higher frame rates, NVIDIA advises that developers incorporate the hybrid rendering pipeline. The ray tracing and the rasterization process can take place simultaneously in Turing GPUs. Essentially, the ray tracing part is handled first with the rays being generated for tracing. Because it is a voluminous processes, there is a scheduler handling the sequence of ray generation. As each traced ray interacts with objects in the scene, the shader is called in to calculate the colours. Once that's done, we have the standard raster model come into the picture and start drawing. Soon enough we have the entire scene calculated and sent to the ROP before it gets displayed on the screen. d